REPO RADAR

satmux: make airplane Wi-Fi usable

Repo: Th0rgal/satmux · MIT license · pure Python 3, no dependencies.

What it does

Inflight Wi-Fi over geostationary satellites (about 620 ms round trip, per the README) often drops connections to sites like github.com. satmux keeps 12 warm TCP connections open on port 8444 to a small Linux server near the satellite ground station, and relays your traffic through them with zlib compression.

Install, step by step

1. On the laptop: `git clone https://github.com/Th0rgal/satmux.git && cd satmux`
2. Make a key: `openssl rand -hex 32 > mux.psk && chmod 600 mux.psk`
3. Rent a small Linux VM near the satellite ground station (README: about $0.01 to $0.03 per hour, Python 3.9+, TCP port 8444 open).
4. Copy `satmux.py` and `mux.psk` to the VM and start the server: `SATMUX_PSK=$(cat mux.psk) python3 satmux.py server --port 8444`
5. On the laptop, start the client: `SATMUX_PSK=$(cat mux.psk) python3 satmux.py client --host <SERVER_IP> --port 8444 --carriers 12`
6. Point apps at the local SOCKS5 proxy `127.0.0.1:1080` or the HTTP proxy `127.0.0.1:8080`. On macOS, `./bin/satmux-macos on <SERVER_IP>` does it for you; `./bin/satmux-macos off` undoes it after landing.

What we tested (and what we did not)

In our cloud sandbox we ran the server and client on localhost, with 4 carriers, and fetched a local page through the SOCKS5 proxy: HTTP 200. We did **not** test on a plane or a satellite link. The flight numbers in the video (3 of 6 sites failed on raw cabin Wi-Fi, 0 of 6 with satmux; 19.13 s to 5.35 s) are the author's own measurements from the README.

3 uses

- A coding session or `git` over cabin Wi-Fi that keeps resetting.
- Loading docs and package registries (PyPI, MDN) on a high-latency link.
- Any slow link where each new connection pays several round trips.

Limits

- The speed of light stays: one satellite hop is still about 580 to 650 ms.
- Your per-seat bandwidth cap still applies.
- It uses your paid cabin Wi-Fi session; it does not bypass captive portals.

Sources: the satmux README and LICENSE (github.com/Th0rgal/satmux); geostationary orbit altitude 35,786 km.

Free guide from Repo Radar. Information only, not financial, legal or medical advice.